

PRACTICA BLUE TEAM OPNSENSE

5.1 Asignación de interfaces (consola)

Tras arrancar instalado verás el **menú de consola** (estas son las 13 opciones reales):

- | | |
|--------------------------------|-----------------------------|
| 0) Logout | 7) Ping host |
| 1) Assign interfaces | 8) Shell |
| 2) Set interface(s) IP address | 9) pfTop |
| 3) Reset the root password | 10) Filter logs |
| 4) Reset to factory defaults | 11) Restart web interface |
| 5) Reboot system | 12) Upgrade from console |
| 6) Halt system | 13) Restore a configuration |

- Si las interfaces no quedaron como quieres, usa **1) Assign interfaces** :
- **LAN** → la NIC de la red interna (la **em0** / **vtnet0** conectada a **lan-blue**).
- **WAN** → la NIC en modo NAT.
- (VLANs: responde **no** .)

Al terminar, OPNsense muestra algo como:

```
WAN (vtnet1) -> v4/DHCP4: 10.0.2.15/24
LAN (vtnet0) -> v4: 192.168.1.1/24
```

```
*** OPNsense.internal: OPNsense 26.1.6_2 (amd64) ***

LAN (em1)      -> v4: 192.168.1.1/24
WAN (em0)      -> v4/DHCP4: 10.0.2.15/24
                v6/DHCP6: fd17:625c:f037:2:a00:27ff:fe6d:a5ca/64

HTTPS: SHA256 02 03 9D B6 6A 8A 30 A1 48 76 82 26 F1 FA DB 3D
           E2 DC 04 40 FC AE D5 C4 FE EC 66 9A FE EB 52 2B

0) Logout          7) Ping host
1) Assign interfaces 8) Shell
2) Set interface IP address 9) pfTop
3) Reset the root password 10) Firewall log
4) Reset to factory defaults 11) Reload all services
5) Power off system 12) Update from console
6) Reboot system    13) Restore a backup

Enter an option: █
```

6. Fase 3 — Acceso web y comprobación del enrutamiento

A OPNsense **se administra desde la LAN** (la WAN, por defecto, bloquea el acceso). Por eso abrimos la consola web **desde Kali o Debian**.

1. Arranca **Kali** y **Debian**. Con DHCP de OPNsense deberían recibir IP automáticamente.
2. En **Debian** (o Kali), comprueba la red:

```
ip a                                # ¿Tienes una IP 192.168.1.x?
ip route                            # default via 192.168.1.1 → la puerta de enlace es OPNsense
cat /etc/resolv.conf                # nameserver 192.168.1.1 → el DNS también es OPNsense
ping -c3 192.168.1.1                # ¿Llegas al firewall?
ping -c3 1.1.1.1                    # ¿Hay Internet por IP? (enrutamiento/NAT)
ping -c3 google.com                 # ¿Hay Internet + DNS? (resolución)
```

```
(kali@kali)~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group def
    qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP g
    roup default qlen 1000
    link/ether 08:00:27:0a:35:d2 brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.188/24 brd 192.168.1.255 scope global dynamic noprefixrout
        e eth0
        valid_lft 86370sec preferred_lft 86370sec
        inet6 fe80::7559:a1ec:4468:cd72/64 scope link noprefixroute
        valid_lft forever preferred_lft forever

(kali@kali)~$ ip route
default via 192.168.1.1 dev eth0 proto dhcp src 192.168.1.188 metric 100
192.168.1.0/24 dev eth0 proto kernel scope link src 192.168.1.188 metric 100

(kali@kali)~$ cat /etc/resolv.conf
# Generated by NetworkManager
search internal
nameserver 192.168.1.1

(kali@kali)~$ ping -c3 192.168.1.1
PING 192.168.1.1 (192.168.1.1) 56(84) bytes of data:
64 bytes from 192.168.1.1: icmp_seq=1 ttl=64 time=0.861 ms
64 bytes from 192.168.1.1: icmp_seq=2 ttl=64 time=0.651 ms
64 bytes from 192.168.1.1: icmp_seq=3 ttl=64 time=2.17 ms

--- 192.168.1.1 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2106ms
rtt min/avg/max/mdev = 0.651/1.228/2.172/0.672 ms

(kali@kali)~$ ping -c3 1.1.1.1
PING 1.1.1.1 (1.1.1.1) 56(84) bytes of data:
64 bytes from 1.1.1.1: icmp_seq=1 ttl=254 time=13.3 ms
64 bytes from 1.1.1.1: icmp_seq=2 ttl=254 time=12.6 ms
64 bytes from 1.1.1.1: icmp_seq=3 ttl=254 time=12.9 ms

--- 1.1.1.1 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2010ms
rtt min/avg/max/mdev = 12.585/12.923/13.275/0.281 ms
```

```
(kali@kali)~$ ping -c3 google.com
PING google.com (216.58.205.78) 56(84) bytes of data:
64 bytes from mil04s25-in-f14.1e100.net (216.58.205.78): icmp_seq=1 ttl=254 t
ime=12.9 ms
64 bytes from mil04s25-in-f14.1e100.net (216.58.205.78): icmp_seq=2 ttl=254 t
ime=12.7 ms
64 bytes from mil04s25-in-f14.1e100.net (216.58.205.78): icmp_seq=3 ttl=254 t
ime=18.8 ms

--- google.com ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 6049ms
rtt min/avg/max/mdev = 12.678/14.781/18.804/2.845 ms
```

Ejercicio 4.1 — Bloquear ping (ICMP) hacia Internet

Objetivo: ver una regla en acción de forma inmediata.

Campo	Valor
Action	Block
Interface	LAN
Direction	in
Protocol	ICMP
Source	LAN net
Destination	any
Log	✓ (marcado)
Description	Lab: bloquear ICMP saliente

Guarda → **Apply changes**. Coloca esta regla **por encima** de la regla "default allow LAN".

1 Action

Block

2 Disabled

☐ Disable this rule

3 Quick

☒ Apply the action immediately on match.

4 Interface

LAN

5 Direction

in

6 TCP/IP Version

IPv4

7 Protocol

ICMP

8 ICMP type

Any

9 Source / Invert

☐ Use this option to invert the sense of the match.

10 Source

LAN network

Advanced

11 Destination / Invert

☐ Use this option to invert the sense of the match.

12 Destination

any

13 Destination port range

from:
any

14 Log

☒ Log packets that are handled by this rule

15 Category

16 Description

Lab: bloquear ICMP saliente

	Protocol	Source	Port	Destination	Port	Gateway	Schedule		Description
Automatically generated rules									
	IPv4 ICMP	LAN net	*	*	*	*	*		Lab: bloquear ICMP saliente
	IPv4 *	LAN net	*	*	*	*	*		Default allow LAN to any rule

Prueba desde Kali/Debian:

```
ping -c3 1.1.1.1 # ahora debe FALLAR (100% packet loss)
curl -I https://example.com # esto SIGUE funcionando (es TCP/443, no ICMP)
```

```
(kali@kali) ~$ ping -c3 1.1.1.1
PING 1.1.1.1 (1.1.1.1) 56(84) bytes of data.
--- 1.1.1.1 ping statistics ---
3 packets transmitted, 0 received, 100% packet loss, time 2045ms

(kali@kali) ~$ curl -I https://example.com
HTTP/2 200
date: Tue, 09 Jun 2026 11:02:08 GMT
content-type: text/html
server: Cloudflare
last-modified: Fri, 05 Jun 2026 20:00:44 GMT
allow: GET, HEAD
accept-ranges: bytes
age: 8595
cf-cache-status: HIT
cf-ray: a08fa46f19e60341-MAD
```

Mira el log en **Firewall > Log Files > Live View**: verás los paquetes ICMP bloqueados con tu descripción.

Pregunta rápida: ¿por qué la web sigue cargando si "no hay Internet"? Porque solo bloqueaste ICMP, no TCP. El firewall filtra por protocolo/puerto, no por "Internet" en abstracto.

Interface	Time	Protocol	Source	Destination	Action	Label
LAN	In 2026-06-09T11:03:17	TCP	192.168.1.188:33390	192.168.1.1:443	pass	anti-lockout rule
WAN	Out 2026-06-09T11:02:08	TCP	10.0.2.15:64241	104.20.23.154:443	pass	let out anything from firewall host itself (force gw)
WAN	Out 2026-06-09T11:02:08	UDP	10.0.2.15:43640	172.64.35.228:53	pass	let out anything from firewall host itself (force gw)
WAN	Out 2026-06-09T11:02:08	UDP	10.0.2.15:30936	172.64.35.228:53	pass	let out anything from firewall host itself (force gw)
WAN	Out 2026-06-09T11:02:08	UDP	10.0.2.15:49940	192.42.93.30:53	pass	let out anything from firewall host itself (force gw)
WAN	Out 2026-06-09T11:02:08	UDP	10.0.2.15:56652	192.48.79.30:53	pass	let out anything from firewall host itself (force gw)
LAN	In 2026-06-09T11:01:38	ICMP	192.168.1.188	1.1.1.1	block	Lab: bloquear ICMP saliente
LAN	In 2026-06-09T11:01:37	ICMP	192.168.1.188	1.1.1.1	block	Lab: bloquear ICMP saliente
LAN	In 2026-06-09T11:01:36	ICMP	192.168.1.188	1.1.1.1	block	Lab: bloquear ICMP saliente

Ejercicio 4.2 – Control por host (segmentación)

Objetivo: bloquear a **Kali** un servicio y dejárselo a **Debian**. Es la base de una DMZ: distintos hosts, distintas políticas.

1. Fija las IPs para identificarlas (mira **ip a** en cada VM; supongamos Kali = **192.168.1.10**).
2. Crea una regla en **LAN**:

Campo	Valor
Action	Reject
Interface	LAN
Protocol	TCP
Source	192.168.1.10 (Kali)
Destination	any
Destination port	HTTPS (443)
Description	Lab: Kali sin HTTPS

Prueba: desde Kali **curl -I https://example.com** → falla al instante (Reject). Desde **Debian** → sigue funcionando. Has aplicado una política **por origen**.

💡 **Pass vs Block vs Reject:** repite el 4.2 cambiando **Reject** por **Block** y observa la diferencia desde Kali: con **Reject** el **curl** falla rápido (recibe un RST); con **Block** se queda **colgado hasta el timeout** (descarte silencioso). Para redes internas se suele preferir **Reject** ; hacia Internet (WAN), **Block** .

⚠️ **Si una regla "no hace efecto":** el firewall mantiene **estados**. Tras crear/editar reglas, resetea con **Firewall > Diagnostics > States** , o cierra la conexión y reintenta.

1 Action

Reject

2 Disabled

☐ Disable this rule

3 Quick

☒ Apply the action immediately on match.

4 Interface

LAN

5 Direction

In

6 TCP/IP Version

IPv4

7 Protocol

TCP

8 Source / Invert

☐ Use this option to invert the sense of the match.

9 Source

Single host or Network

192.168.1.188

Advanced

10 Destination / Invert

☐ Use this option to invert the sense of the match.

11 Destination

any

12 Destination port range

from:

HTTPS (443)

13 Log

☐ Log packets that are handled by this rule

14 Category

15 Description

Lab: Kali sin HTTPS

☐	Protocol	Source	Port	Destination	Port	Gateway	Schedule		Description
☐									Automatically generated rules
☐		192.168.1.188	*	*	HTTPS (443)	*	*		Lab: Kali sin HTTPS
☐		LAN net	*	*	*	*	*		Lab: bloquear ICMP saliente
☐		LAN net	*	*	*	*	*		Default allow LAN to any rule

```
(kali@kali)~$ curl -i https://example.com
curl: (7) Failed to connect to example.com port 443 after 834 ms: Could not connect to server
```

no tenia curl instalado, use timeout:

```
debian@debian:~$ timeout 3 bash -c '</dev/tcp/1.1.1.1/443' && echo "Puerto ABIERTO" || echo "Puerto CERRADO o BLOQUEADO"
Puerto ABIERTO
```

8. Fase 5 — Bloquear sitios web (la forma correcta)

El concepto que casi todo el mundo confunde

Un firewall de paquetes filtra por **IP y puerto** (capa 3/4). **No entiende de dominios**. Para bloquear `ejemplo.com` necesitas algo que trabaje a **nivel de aplicación/DNS (capa 7)** — exactamente lo que decía la lectura de **servidores proxy**.

Tienes dos vías limpias en OPNsense:

Método A (recomendado) — Bloqueo por DNS con Unbound *Blocklists*

OPNsense resuelve el DNS de tus clientes (es su DNS), así que puede "agujerear" (sinkhole) los dominios que no quieras.

Menú: **Services > Unbound DNS > Blocklists**.

Campo	Qué hacer
Enable	✓
Wildcard Domains	Añade un dominio de prueba, p. ej. <code>example.com</code> (bloquea también todos sus subdominios). <i>No se puede bloquear un TLD entero como <code>com</code>.</i>
Blocklist Domains	Lista de dominios exactos a bloquear (sin comodín, sin regex).
Type of DNSBL	(Opcional) marca listas predefinidas: AdAway, AdGuard, OISD, EasyList, Abuse.ch ThreatFox... para bloquear publicidad/malware en masa.
Allowlist Domains	Excepciones (lista blanca).
Return NXDOMAIN	Devuelve NXDOMAIN en vez de <code>0.0.0.0</code> (útil para dispositivos quisquillosos).

Guarda → **Apply**. (El cambio tarda hasta ~1 min en cargarse; Unbound no se reinicia.)

Prueba desde Kali/Debian:

```
nslookup example.com 192.168.1.1    # → debería responder 0.0.0.0 (o NXDOMAIN)
# o con dig (instala dnsutils si no lo tienes: sudo apt install dnsutils):
dig example.com @192.168.1.1
```

Y en el navegador: `http://example.com` ya no carga.

Edit Blocklist

advanced mode
full help

1
Enable
☒

1
Type of DNSBL
Abuse.ch - ThreatFox IOC database, AdGuard List, |

Clear All
Select All

1
URLs of Blocklists

Clear All
Copy
Paste
Text

1
Allowlist Domains

Clear All
Copy
Paste
Text

1
Blocklist Domains
example.com

Clear All
Copy
Paste
Text

1
Wildcard Domains
example.com

Clear All
Copy
Paste
Text

1
Source Net(s)

Clear All
Copy
Paste
Text

1
Cache TTL
72000

1
Destination Address

1
Return NXDOMAIN
☒

1
Description
Lab: bloquear example.com

Cancel
Save

borrar caché para que funcione:

```
(kali@kali)-[~]
$ sudo systemd-resolve --flush-caches 2>/dev/null || sudo systemctl restart systemd-resolved 2>/dev/null
[sudo] contraseña para kali:
```

```
(kali@kali)-[~]
$ nslookup example.com 192.168.1.1
Server:      192.168.1.1
Address:     192.168.1.1#53

** server can't find example.com: NXDOMAIN
```

9. Fase 6 — "Algo más complejo" (elige uno o varios)

9.1 Forzar el uso del DNS del firewall (anti-bypass)

Problema real: un usuario listo configura **8.8.8.8** a mano y se **salta** tu bloqueo por DNS. Solución de Blue Team: **redirigir y/o bloquear** todo el DNS saliente.

- Redirección (NAT):** Firewall > NAT > Port Forward → en **LAN**, Protocolo **TCP/UDP**, **Destination port = DNS (53)**, **Redirect target IP = 127.0.0.1**, **Redirect target port = 53**. Esto reconduce cualquier consulta DNS al propio Unbound.
- Bloqueo de fuga:** una regla **Block** en LAN para **Destination port 53** hacia destinos que **no** sean OPNsense.

Prueba: desde Kali `nslookup example.com 8.8.8.8` → debería seguir devolviendo tu sinkhole (no el resultado real de Google), demostrando que ya **no pueden esquivar** el filtro.

Es la misma técnica recomendada por OPNsense cuando usas DNS-over-TLS: bloquear el 53 saliente y redirigir a **127.0.0.1:53**.

rules destination NAT:

Edit Rule

advanced mode

full help

Organization

Disabled

Categories

Description

Interface

Version

Protocol

Source (advanced)

Destination

Invert Destination

Destination Address

Destination Port

Translation

Redirect Target IP

Redirect Target Port

☐		LAN	*	TCP	*	*	LAN address	443	*	*	Anti-Lockout Rule
☐		LAN	*	TCP	*	*	LAN address	80	*	*	Anti-Lockout Rule
☐		LAN	*	TCP/UDP	*	*	LAN network	domain	127.0.0.1	domain	Lab: anti-bypass DNS

☐	Protocol		Source	Port	Destination	Port	Gateway	Schedule	Description		
☐	Automatically generated rules										
☐		IPv4 TCP/UDP	LAN net	*	LAN address	DOMAIN (53)	*	*	Lab: bloquear fuga DNS saliente		
☐		IPv4 TCP	192.168.1.188	*	*	HTTPS (443)	*	*	Lab: Kali sin HTTPS		
☐		IPv4 ICMP	LAN net	*	*	*	*	*	Lab: bloquear ICMP saliente		

```
(kali@kali) ~  
$ nslookup example.com 8.8.8.8  
;; communications error to 8.8.8.8#53: timed out  
;; communications error to 8.8.8.8#53: timed out  
;; communications error to 8.8.8.8#53: timed out  
;; no servers could be reached
```

Interface	Time	Protocol	Source	Destination	Action	Label
LAN	In	2026-06-09T12:28:01	TCP	192.168.1.188-41474	192.168.1.1-443	pass anti-lockout rule
LAN	In	2026-06-09T12:25:46	UDP	192.168.1.188-43752	8.8.8.8-53	block Lab: bloquear fuga DNS saliente
LAN	In	2026-06-09T12:25:41	UDP	192.168.1.188-51775	8.8.8.8-53	block Lab: bloquear fuga DNS saliente
LAN	In	2026-06-09T12:25:36	UDP	192.168.1.188-47984	8.8.8.8-53	block Lab: bloquear fuga DNS saliente

rules lan:

Action

Block

Disabled

Quick

Interface

Direction

TCP/IP Version

Protocol

Source / Invert

Source

Destination / Invert

Destination

Destination port range

Log

Category